

USB Gamepad Standalone Control

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1. Course Content

- Use a USB handle connected directly to the robot control board to control the robot's vehicle controller

2. Brief Introduction to the Working Principle

Unlike Section 4.2, which processes handle signals through a chain of ROS2 nodes, the data path in this mode forms a closed loop entirely on the control board:

USB handle → control board USB port → firmware parses handle data → directly drives motor/servo/buzzer

- After power-up, the firmware automatically detects whether a handle is plugged into the USB port
- Once a handle is detected, it autonomously handles joystick reading, button parsing, and command generation

3. Control Priority and Switching

The USB standalone handle and ROS2 topic control can be online at the same time; **the USB standalone handle has higher priority**:

Control signal source	Priority	Effective condition
USB standalone handle (this section)	High	when the handle is plugged into the USB port and has valid input
ROS2 <code>/cmd_vel</code> (4.1/4.2)	Low	Automatically takes over when there is no USB handle input

- When the USB handle has valid joystick input → the firmware uses the USB handle commands and ignores `/cmd_vel`
- When the USB handle has no input (joystick centered or unplugged) → the firmware automatically switches, and `/cmd_vel` takes over

4. Operating Instructions

The button functions are the same as the ROS2 handle control in Section 4.2:

SELECT (Only valid for ROS2 version of the handle control program)

Single press: Enable/disable handle ROS2 program control

Arrow keys

Up/Down: Forward/Backward

Left/Right: Left/Right

START: Press the
buzzer to sound,
Release the buzzer
no sound

(For robot car with PTZ)

Y: PTZ upwards

A: PTZ downwards

X: PTZ turn left

B: PTZ turn right



Left remote control
Up/Down: forward/backward

Right remote control
Left/Right: Turn left/right

R1: Angular velocity acceleration

L1: Linear velocity acceleration



R2: Angular velocity deceleration

L2: Linear velocity deceleration

Handle connection status: The indicator light is constantly on,
indicating that the connected successfully

4.1 Movement Control

Input	Function
Left stick up/down	Forward / Backward
Right stick left/right	Turn left / Turn right
D-pad	Digital fallback when the joystick is idle

4.2 Speed Levels

A fixed 5-level speed table (G2 is the default level):

Level	Linear speed (m/s)	Angular speed (rad/s)
G1	0.20	0.50
G2 (default)	0.30	1.50
G3	0.50	2.50
G4	0.75	3.75
G5	1.00	5.00

Key	Function
L1	Increase linear speed by one level (0→1 rising edge trigger)
L2	Decrease linear speed by one level (0→1 rising edge trigger)
R1	Increase angular speed by one level (0→1 rising edge trigger)
R2	Decrease angular speed by one level (0→1 rising edge trigger)

4.3 Handle Control

Key	Function
X	Servo S1 forward
B	Servo S1 reverse
Y	Servo S2 reverse
A	Servo S2 forward

4.4 Buzzer Feedback

Trigger condition	Behavior
START (OPTIONS) held	Beeps continuously
Gear shift succeeded (up or down one level)	Beeps once briefly
Gear shift hits a boundary (G1/G5)	Beeps twice briefly